Date:_____ Day:_____

Question 1: (IPO – Chart)

Input	Process	Output
• Rows • Seats per row • Status of every seat (0 or 1) • Group size K	• Store the seats in 2D-array • validate K (more than 0 & < seats per row) • Count free seats in row • Add up to total free seats. • Start from the middle (row/2) • Check the other rows by distance from the middle. • On a tie, check smaller row first. • Find first K consecutive free seats • Mark those seats as 1.	• Seating plan (1 & X) • Free seats per row • Total free seats • Allocated row & Seat numbers. • Update plan & Remaining seats. • "No seats" if no block is found.

Question 2: (IPO Chart)

Input	Process	Output
• Number n	• loop from 1 – n • Find sum of proper divisors • Find sum of divisors of that sum • check that it gives back the 1st no: • Skip no: is equal to their own sum (6, 28) • Keep the smaller no: first • Print each pair once. • Count the pair	• list of amicable pairs. • Total pairs • "None" if no pair is found.

Teacher's Signature:_____

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Date:_____ Day:_____

Question 3: (TPO Chart)

Input	Process	Output
• No: of Sensors • No: of daily reading • All temp: reading.	• Store the reading in 2D-array. • For each Sensor calculate - Total • Average • Highshest • Critical cant (40 or above) • warning pt (35–39) - Longest streak at 35–above • Find Sensor with highest average	• Repeat for each Sensor. • Average with two decimal place. • Sensor with the highest average.

Question 4: (TPO Chart)

Input	Process	Output
• No: of days (1–7) • No: of order each day. • Amnt of each Order	• Re-ask until the days are valid. ael until amnt > 0. • Each day calculate ⇒ total, average, highest • Store daily totals in 1D array • Adel up festival total • Find the earliest day with highest sales	• Daily (total, avg highest order • Fes: total Sales • Best day and its Sales • Bar Chart

Teacher's Signatur

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Date: Day:

Question 5: (IPO Chart)

Input	Process	Output
• Menu Choice • Mem-Name ID, art type • Book ID • No: of days • Membership Code.	• Show the menu in loop and use a Switch. • Register the member • Validate the book ID and availability • Allow aman: 3 books • Record borrow & reduce copies • On, return, check the book was borrowed. • Increase copies on return • Find late days (14 or 30) • Calc: the fine (Std: 10, faculty 20 per extra day. • If the code is LIB2026 give 50% wavier. • Remove the book from the borrowed List. • Update the fine record	• Menu • Book table • Borrow and return messages. • Fine and final payable amount. • Member record.